

ARSHAD SYEED

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SUMMARY

B.Tech Electronics and Communication Engineering student (CGPA 8.52/10, Class of 2027) building production-style, full-stack AI software: Python/FastAPI backends, React/TypeScript frontends, and LLM-powered systems spanning Retrieval-Augmented Generation, multi-provider LLM routing (Groq, Gemini, Ollama), and real-time WebSocket applications. Experience shipping tested, CI/CD-integrated software (240+ automated tests) and two hands-on engineering internships in signaling systems and hardware diagnostics. Google AI Professional Certificate holder developing product-engineering skills across the stack.

EDUCATION

Vel Tech Rangarajan Dr. Sagunthala R&D Institute

B.Tech, Electronics and Communication Engineering

CGPA: 8.52/10 | Class XII: 82.6% | Class X: 99.7%

2023 – 2027

Chennai, India

EXPERIENCE

Signals and Telecommunication Intern

South Central Railways

Jul 2025 – Sep 2025

Andhra Pradesh, India

- Analyzed fault-detection workflows and interlocking logic for a Network Failure Detection System across 4 operational departments, improving alert-generation reliability for safety-critical infrastructure.
- Audited real-time signaling and monitoring pipelines, identifying systemic gaps in live railway control architecture.

Electronics Intern

Frontline Electronics Pvt. Ltd.

Jun 2025

Telangana, India

- Diagnosed hardware faults in aviation warning lighting systems via PCB-level root-cause analysis; performed PCB fabrication, SMT inspection, and quality-control workflows.

PROJECTS

JASS AI | AI-Powered Personal Assistant Platform

Mar 2026 – Present

Python, FastAPI, React, FAISS, RAG, WebSockets, JWT, SQLAlchemy, Groq, Gemini, Ollama, Whisper

- Engineering a hybrid Retrieval-Augmented Generation pipeline combining FAISS dense vector search with BM25 sparse retrieval, query expansion, and context compression for long-term memory.
- Building a JWT-secured FastAPI backend with SQLAlchemy and WebSockets for real-time communication, plus a multi-provider LLM routing layer across Groq, Gemini, and Ollama with real-time voice input via Whisper.

StudyFlow AI | AI-Powered Student Productivity Platform

Jun 2026 – Aug 2026

Electron, JavaScript, React, TypeScript, Vite, Python, FastAPI, SQLite, PostgreSQL, GitHub Actions

- Built and shipped a full-stack, cross-platform desktop productivity app (Electron + JavaScript, FastAPI backend, SQLite/PostgreSQL) covering task management, goal roadmaps, study planning, focus mode, exam prep, notes, analytics, wellness tracking, and an AI study coach.
- Implemented a multi-provider AI management system for personalized study assistance and a responsive marketing site (React, TypeScript, Vite).
- Added CI workflows and automated testing across backend and desktop components, reaching 240+ passing tests; open-sourced the project with full documentation, contribution guidelines, and a security policy.

AI Email Assistant

May 2026 – Jun 2026

Python, Gmail API, OpenRouter, SQLite, Streamlit, Telegram Bot API, OAuth 2.0

- Automated email classification into 5 categories, summarization, priority scoring, and reply drafting using Gmail OAuth 2.0 and OpenRouter LLMs.
- Engineered a human-in-the-loop decision layer routing personal/commitment emails through a Telegram approval workflow (Send, Edit, or Ignore), with a Streamlit analytics dashboard and SQLite audit logging.

Line Detection & Obstacle Avoidance Robot Car

Feb 2026 – Mar 2026

- Built an autonomous ESP32 robot (IR + ultrasonic sensors, C++) with motor control and sensor-fusion logic for dynamic path correction.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++

Backend & APIs: FastAPI, REST APIs, WebSockets, Node.js, JWT, OAuth 2.0

AI / LLM Systems: RAG, FAISS, BM25, Vector Databases, LLM APIs (Groq, Gemini, Ollama, OpenRouter), Prompt Engineering, Generative AI

Frontend & Databases: React, TypeScript, HTML/CSS | PostgreSQL, SQLite, MySQL, SQLAlchemy

Developer Tools: Git, GitHub, GitHub Actions, CI/CD, VS Code, Streamlit, Electron

Other Domains: Embedded Systems (C++, Arduino IDE, ESP32), Signal & Telecommunication, LTspice

CERTIFICATIONS & ACHIEVEMENTS

- Google AI Professional Certificate — Google via Coursera, May 2026
- JavaScript Certification — Udemy, Oct 2025 | MySQL Certification — Udemy, Dec 2025
- HackVega 2.0 — Final Round Qualifier, nationwide hackathon (47,000+ registrations), by MyCareernet & HirePro
- IEEEExtreme 18.0 — Global Rank 1788, India Rank 556, University Rank 15 (IEEE's 24-hour international programming contest), Oct 2024